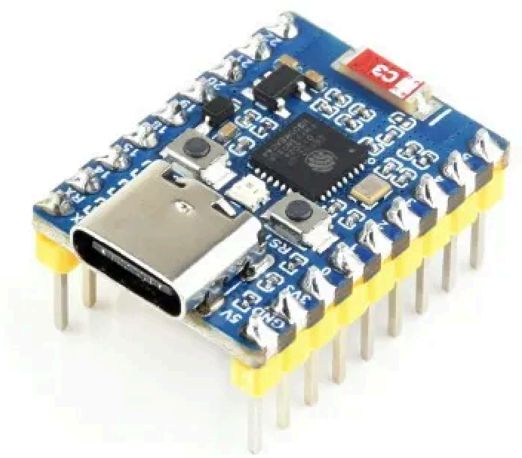
ESP32-C6-Zero

ESP32-C6-Zero (Standard Version) | **ESP32-C6-Zero-M (Header Version)** | **ESP32-C6-Zero-Basic-Kit (Basic Kit)**



The ESP32-C6-Zero is a compact microcontroller development board that integrates a variety of digital interfaces.

On the hardware side, it uses the ESP32-C6FH8 chip, featuring a 32-bit RISC-V high-performance processor (HP Core) running at up to 160 MHz and a 32-bit low-power coprocessor (LP Core) running at up to 20 MHz. The chip comes with 8 MB Flash, 320 KB ROM, 512 KB HP SRAM, and 16 KB LP SRAM, making it easy to expand various peripherals.

On the software side, it supports development with the ESP-IDF framework or Arduino IDE, allowing quick prototyping and deployment in actual products.

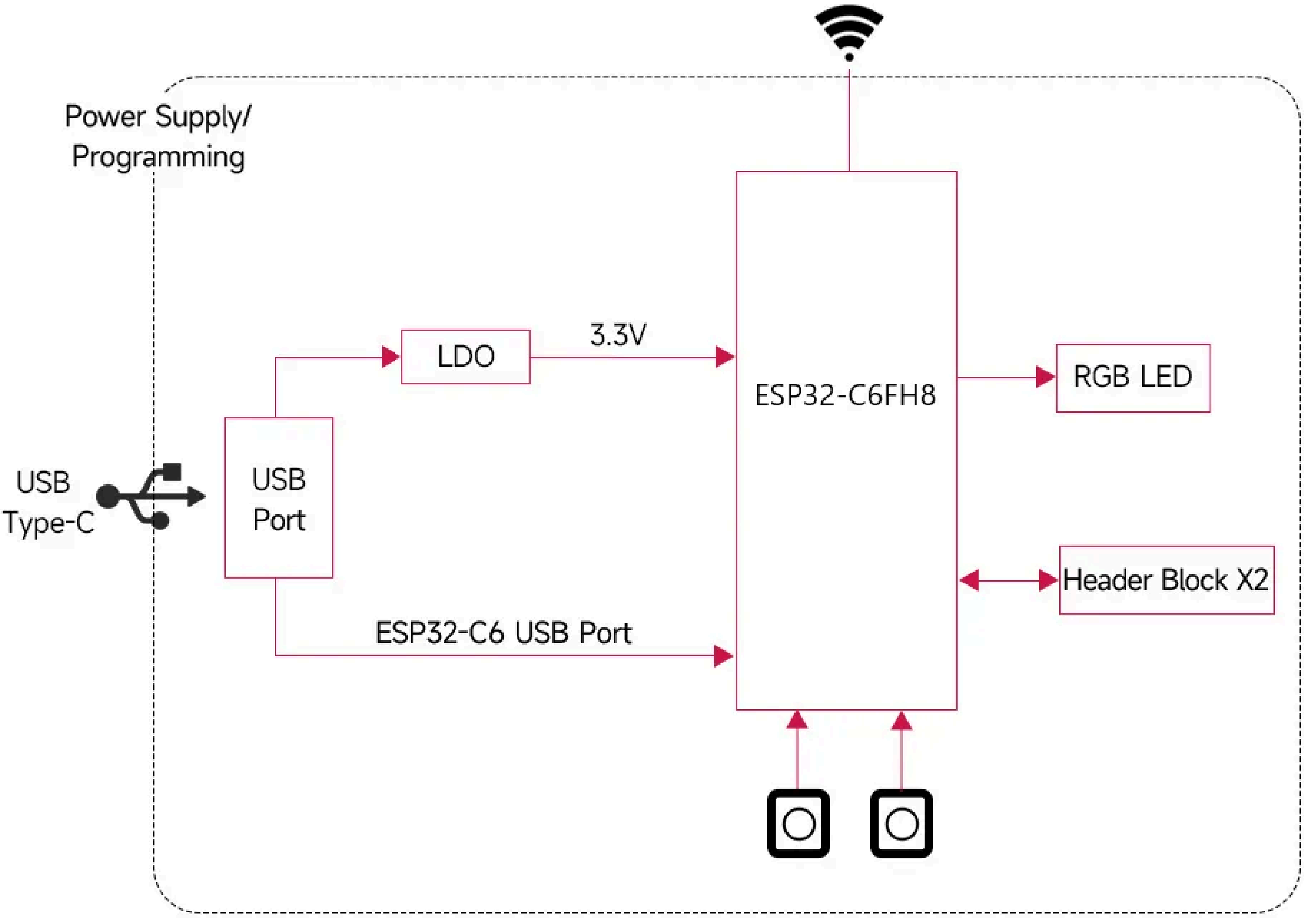
The ESP32-C6-Zero-Basic-Kit includes the ESP32-C6-Zero-M (Header Version) and the ESP32-XX-Basic-Kit-Acce (Kit), featuring a breadboard, wires, LEDs, resistors, a 1.5inch OLED, and other modules. It provides the core hardware needed to learn the Waveshare Arduino Getting Started Tutorial and the Waveshare MicroPython Getting Started Tutorial, making it suitable for quickly getting started with Arduino, MicroPython, and ESP-IDF programming. The related example code and wiring diagrams can be downloaded here: GitHub Repository

SKU	Product
27035	ESP32-C6-Zero
26976	ESP32-C6-Zero-M
33357	ESP32-C6-Zero-Basic-Kit

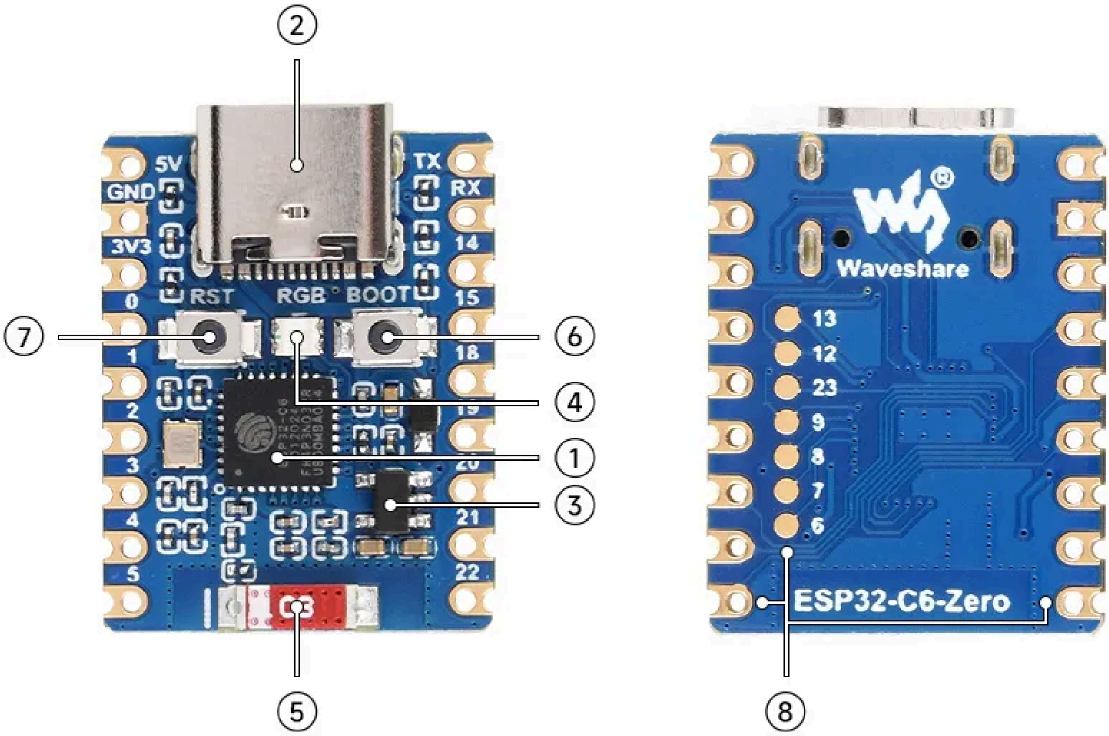
Features

- Powered by the ESP32-C6FH8 chip, integrating a high-performance and a low-power dual-core 32-bit RISC-V processor running at up to 160 MHz and 20 MHz respectively
- Built-in 320 KB ROM, 512 KB HP SRAM, 16 KB LP SRAM, and 8 MB Flash storage
- Integrated Wi-Fi 6, Bluetooth 5, and IEEE 802.15.4 (Zigbee 3.0 and Thread) wireless communication with excellent RF performance
- USB Type-C Port, supports plugging in either orientation
- Rich peripheral interfaces with outstanding compatibility and expandability
- Castellated module design for direct soldering onto carrier boards
- Supports multiple low-power modes, allowing flexible adjustment of communication range, data rate, and power consumption for various application scenarios

Functional Block Diagram

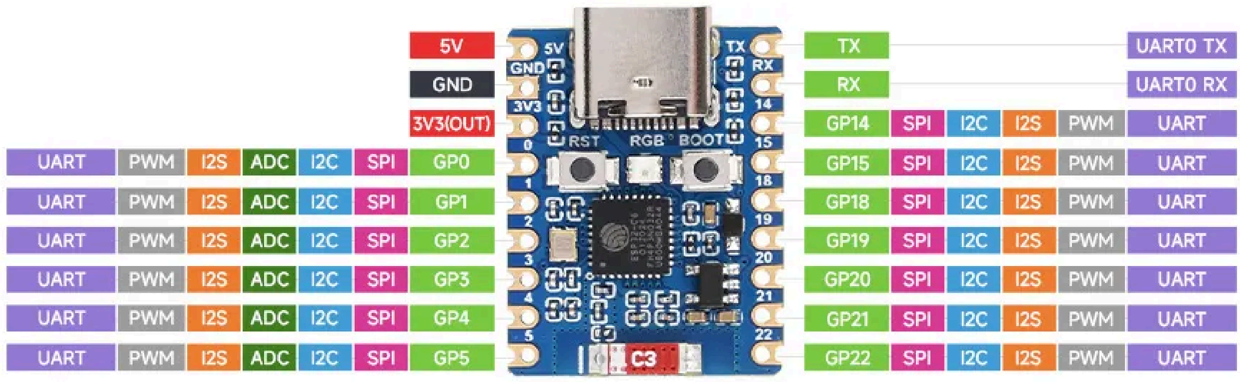


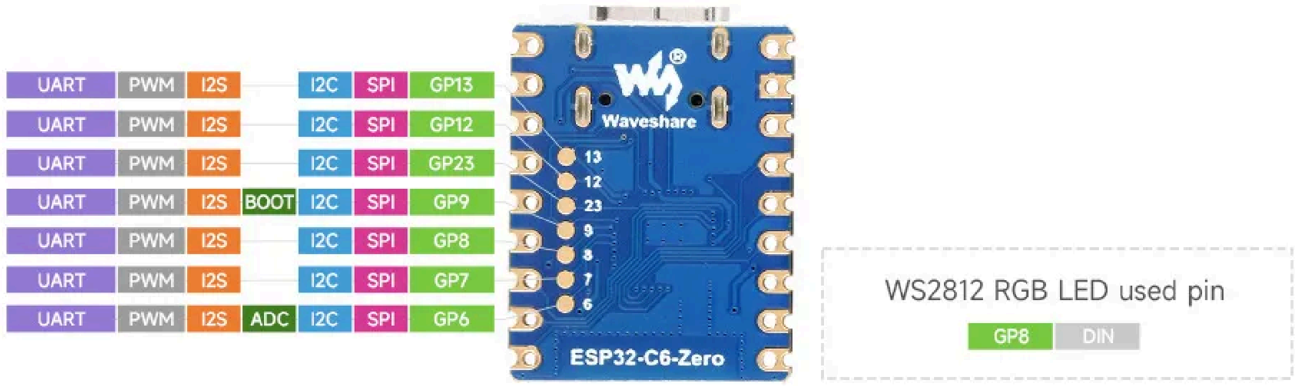
Resource Interfaces



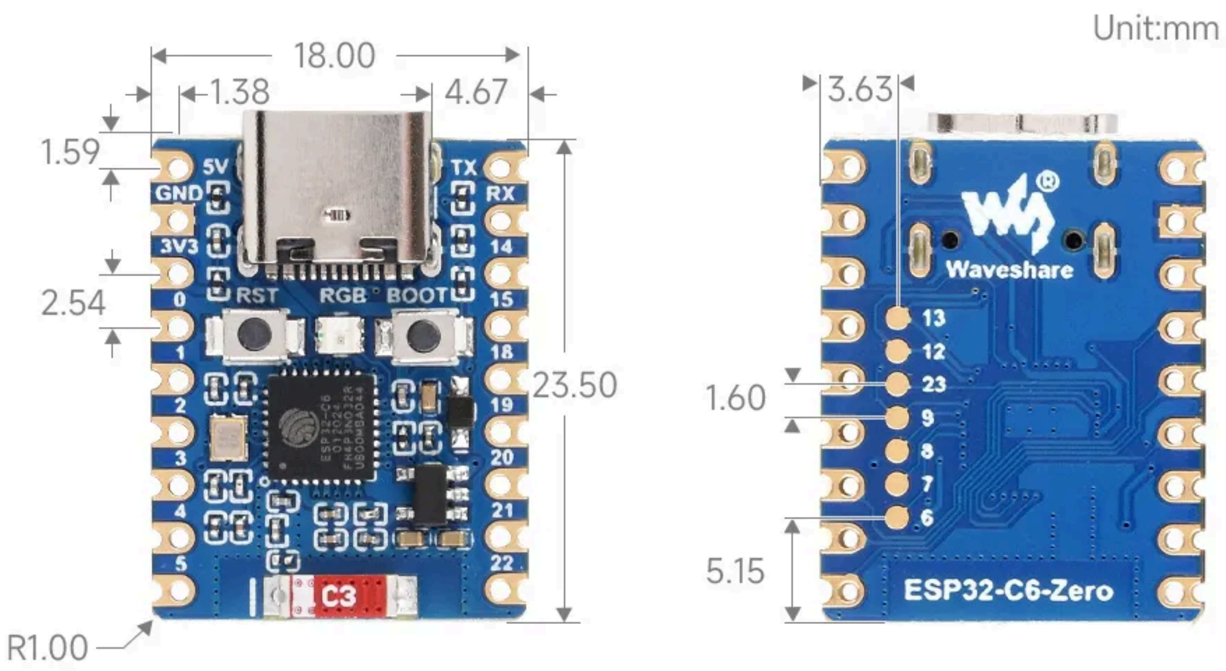
1. **ESP32-C6FH8** Equipped with a RISC-V 32-bit dual-core processor (HP core up to 160 MHz, LP core up to 20 MHz)
2. **USB Type-C Port**: For program download and debugging
3. **ME6217C33M5G**: Low dropout linear regulator, maximum output current 800mA
4. **WS2812 Colorful RGB LED**
5. **2.4G Ceramic Antenna**
6. **BOOT Button**: Hold this button and press the RESET button to enter download mode
7. **RESET Button**
8. **ESP32-C6FH8 Pin Headers**

Pinout Definition


5V
GND
3V3(OUT)
UART PWM I2S ADC I2C SPI GP0
UART PWM I2S ADC I2C SPI GP1
UART PWM I2S ADC I2C SPI GP2
UART PWM I2S ADC I2C SPI GP3
UART PWM I2S ADC I2C SPI GP4
UART PWM I2S ADC I2C SPI GP5
TX UART0 TX
RX UART0 RX
GP14 SPI I2C I2S PWM UART
GP15 SPI I2C I2S PWM UART
GP18 SPI I2C I2S PWM UART
GP19 SPI I2C I2S PWM UART
GP20 SPI I2C I2S PWM UART
GP21 SPI I2C I2S PWM UART
GP22 SPI I2C I2S PWM UART
RST RGB BOOT
C3


UART PWM I2S I2C SPI GP13
UART PWM I2S I2C SPI GP12
UART PWM I2S I2C SPI GP23
UART PWM I2S BOOT I2C SPI GP9
UART PWM I2S I2C SPI GP8
UART PWM I2S I2C SPI GP7
UART PWM I2S ADC I2C SPI GP6
Waveshare
ESP32-C6-Zero
WS2812 RGB LED used pin
GP8 DIN

Dimensions


Unit:mm
18.00
1.38
4.67
1.59
2.54
23.50
R1.00
3.63
1.60
5.15
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